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H. Cory and O. Skorka

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Abstract

The coupling between two identical square dielectric waveguides with parallel axes has been studied for various configurations, in which one of the guides was arbitrarily rotated with respect to the other. In one case, the first guide was kept fixed while the second guide was rotated from 0° to 180°. It was found that the coupling coefficient was maximal when the inclination angle was 0° or 180°, and vanished when it was 90°. In another case, one corner of the first guide faced one corner of the second guide. It was found that the coupling coefficient was less than half the maximal value obtained for the previous configuration. This study has a practical aspect: it proposes a possible way to select an appropriate coupling in order to satisfy a given engineering design.

1. Introduction

Dielectric waveguides have been studied for their potential applications in the transmission of millimeter, submillimeter and optical waves. Parallel coupled guides which can be used as directional couplers, as well as non-parallel coupled guides whose axes are coplanar [1]-[3], have been investigated. The coupling characteristics of two skew rectangular dielectric waveguides have been studied in [4]. A subsequent analysis, based on a different coupled mode theory [5], has been used [6] and has given improved results, especially in the case of strong coupling.

The purpose of this paper is to study the coupling between two rectangular dielectric waveguides whose axes are parallel, arbitrarily rotated in various configurations. The method is general, but for the sake of simplicity, we have assumed the two guides to have the same square cross-section, the same permittivity, and the same excitation (a dominant mode propagation).

2. The Electromagnetic Fields and the Coupling Coefficient

We have studied dielectric waveguides in air and used the well-known dielectric constant method [7]-[10] to calculate the longitudinal propagation coefficient β in the z-direction as well as the transverse propagation coefficients k_x and k_y in the x- and y-directions respectively, for the dominant TE_{11} mode. We have assumed that the field components of the electromagnetic wave have a time-dependence $e^{j\omega t}$ and that they propagate in the z-direction with an $e^{-j\beta z}$ variation. Explicit expressions for these components can be found in [7] in the nine regions described in Fig. 1.

The coupling coefficient has been calculated by a method due to Arnaud [14], which has been used successfully to find the transverse coupling between adjacent guides of various shapes and composition [11]-[13]. In this study, one of the guides (the lower guide a) remains fixed in space while the other one (the upper guide b) rotates about an axis parallel to $\hat{\mathbf{z}}$ and passing through its center (see Fig. 2). The cross-section of the identical guides is a square of side c and the distance between their centers is $c + d_0$. Their dielectric constant is ε.

The general expression for the coupling coefficient is given by [14]

$$\Delta\beta = \tilde{\beta} - \beta = C/P \,,$$

where β is the propagation coefficient of an isolated guide (a or b) while $\tilde{\beta}$ is the propagation coefficient of the two adjacent guides (a and b).

The term C is given as follows:

$$C = \frac{1}{2} \oint_L (\mathbf{E}_a \times \mathbf{H}_b^+ - \mathbf{E}_b^+ \times \mathbf{H}_a) \cdot \mathbf{dl} \, .$$

The indices a and b are related to the lower and to the upper guide respectively. The adjoint fields $\mathbf{E}^+$ and $\mathbf{H}^+$ are defined by $\mathbf{E}^+ = (E_x, E_y - E_z)$ and $\mathbf{H}^+ = (-H_x, -H_y, H_z)$. The line integral over L encircles one of the guides (a in our case). It is taken along the x-axis and closes at infinity. Since at infinity the evanescent waves vanish, only the x-axis contribution is to be taken into consideration so that $\mathbf{dl} = \hat{\mathbf{y}}\, dx$ and

$$C = \frac{1}{2} \int_{-\infty}^{\infty} (E_{bx}\, H_{az} + E_{bz}\, H_{xa} - E_{ax}\, H_{bz} - E_{az}\, H_{bx})\, dx \, .$$

In our case $E_z = 0$ since we are dealing with the TEE mode.

The term P is given as follows:

$$P = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (\mathbf{E}_a \times \mathbf{H}_a) \cdot \hat{\mathbf{z}}\, dx\, dy \, .$$

The surface integral encompasses all the x-y plane. It is given by

$$P = \frac{\beta}{\omega \varepsilon_0} A_0^2\, f(c, \omega, \varepsilon_r) \, ,$$

where f is given in Appendix 1 and A_0 is the amplitude of the wave of guide a.

3. Coupling Calculations and Graphs

Throughout this paper we shall assume some physical quantities to have the following values: $f = 50 \times 10^9 H_z$, $c = 4 \times 10^{-3}$m, $c + d_0 = 6.2 \times 10^{-3}$m, $\varepsilon = 2 \cdot 01\, \varepsilon_0$, $\mu = \mu_0$. The value of P will always be equal to 1, thus determining the value of A_0. The distance between the centers of the guides, i.e., $c + d_0$, will remain fixed for any configuration.

3.1. The upper guide rotates

- **Guide rotation from 0° to 90°**

  It can be seen from Fig. 3 that the x-axis has to be divided into six zones of integration. To each one of these six zones corresponds a different set of expressions for the components of the electromagnetic fields, according to the mapping of the regions from 1 to 9 given in Fig. 1. The data corresponding to this case are summarized in Table 1.

x-axis zone	Begins at	Ends at	Region of the x-y plane where the fields are calculated for guide a	Region of the x-y plane where the fields are calculated for guide b
1	$-\infty$	C	8	9
2	C	E	8	5
3	E	B	4	5
4	B	F	4	2
5	F	H	6	2
6	H	∞	6	6

Table 1.

- **Guide rotation from 90° to 180°**

  It can be seen from Fig. 4 that the x-axis is once more divided into six zones of integration. The data corresponding to this case are summarized in Table 2.

x-axis zone	Begins at	Ends at	Region of the x-y plane where the fields are calculated for guide a	Region of the x-y plane where the fields are calculated for guide b
1	$-\infty$	C	8	7
2	C	E	8	2
3	E	B	4	2
4	B	F	4	4
5	F	H	6	4
6	H	∞	6	8

Table 2.

- **Coupling variation with inclination angle in the two preceding cases**

  The integrand of C is given in Fig. 5 as a function of the distance along the x-axis for various inclination angles θ, and the coupling coefficient $\Delta\beta = C/P$ is given in Fig. 6 as a function of the inclination angle θ. It should be noted that the x-axis, along which the integration is performed, always contains the lower corner of guide b (point B). This axis approaches guide a when the rotation is performed from 0° to 45° but recedes from it when it is performed from 45° to 90°, so that the interaction between guides a and b is strongest in the neighbourhood of this corner for $\theta = 45°$. Moreover, it can be seen from Fig. 5 that the integrand of C has specific symmetry properties with respect to the origin of coordinates so that the coupling coefficients $\Delta\beta$ are identical for θ and $180° - \theta$. The coupling coefficient vanishes for $\theta = 90°$ ($\beta = \tilde{\beta}$) and is maximal for $\theta = 0°$ and for $\theta = 180°$. In this case $\Delta\beta = 23.63$.

We have proposed in this section a configuration which could provide any required coupling between a maximum value and zero.

3.2. The guides corners face each other

The two guides are now positioned so that the lower corner of guide b faces the upper corner of guide a. There are two different cases: in the first one, the axes of the two guides are mutually parallel (see Fig. 7) while in the second one, these axes are mutually perpendicular (see Fig. 8). It should be noted that for this configuration the path of integration along the x-axis is symmetrical with respect to the two guides and does not pass through the lower corner of guide b as in the previous configuration (see section 3.1). In Tables 3 and 4 are summarized the data corresponding to the two cases treated in this section.

x-axis zone	Begins at	Ends at	Region of the x-y plane where the fields are calculated for guide a	Region of the x-y plane where the fields are calculated for guide b
1	$-\infty$	B	9	9
2	B	C	3	5
3	C	D	8	7
4	D	E	4	2
5	E	∞	6	6

Table 3.

x-axis zone	Begins at	Ends at	Region of the x-y plane where the fields are calculated for guide a	Region of the x-y plane where the fields are calculated for guide b
1	$-\infty$	B	9	7
2	B	C	3	2
3	C	D	8	6
4	D	E	4	4
5	E	∞	6	8

Table 4.

The integrand of C is given in Fig. 9 as a function of the distance along the x-axis for the two configurations considered in this section. It can be seen from the graphs that there is only a slight difference in the value of the ordinate for the two cases. Moreover, $\Delta\beta = 9.11$ for the parallel case and $\Delta\beta = 10.22$ for the perpendicular case. This value is less than half the maximal value obtained for the configuration where the guides are parallel or anti-parallel, i.e. 23.63. The reason is that in these cases, the common neighbourhood of the guides is of larger extent and the fields are stronger there, than in the cases where their corners face each other.

We have shown in this section that by suitably choosing the relative orientation of the two guides. i.e., by lessening the fields in the region where the guides are close, the coupling can be lowered. thus avoiding the undesirable effect of crosstalking.

4. Conclusion

The coupling of two identical square dielectric waveguides, whose sides are mutually inclined, has been studied in different configurations. In the first configuration the lower guide has been kept fixed while the upper one has been rotated. It has been found that the coupling coefficient is highest when the inclination angle of the guides is 0° or 180° and vanishes when the inclination angle is 90°. In the second configuration the two guides have been positioned so that the lower corner of the upper one faces the upper corner of the lower one. It has been found that the values of the coupling coefficients are not very different if the axes of the guides are parallel or perpendicular. The coupling obtained for this configuration is less than half the maximal coupling obtained for the configuration where the guides are parallel or anti-parallel. The above study has thus suggested some configurations which could provide the amount of coupling required to satisfy a given engineering design.

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Appendix 1

The value of P

The expression for P for the identical square waveguides of side c is given as follows [13]. It is assumed that $A_0 = B_0$. Following normalization, $P = 1$.

$$f[c,\omega,\varepsilon_r] = \frac{c^2\,\varepsilon_r}{4\,\varepsilon_{rex}} \cdot \left[\frac{\sin k_x c}{k_x\,c} + 1\right] \cdot \left[\frac{\sin k_y\,c}{k_y\,c} + 1\right]$$
$$+ \frac{c\cos^2 k_x\,\frac{c}{2}}{2\varepsilon_{rex}^2\,\alpha_x} \cdot \left[\frac{\sin k_y c}{k_y\,c} + 1\right] + \frac{c\cos^2\,k_y\,\frac{c}{2}}{2\,\alpha_y} \cdot \left[\frac{\sin k_x\,c}{k_x\,c} + 1\right]$$
$$+ \frac{\left(\cos^2 k_x\,\frac{c}{2}\right) \cdot \left(\cos^2 k_y\,\frac{c}{2}\right)}{\alpha_x \alpha_y}\,.$$

Appendix 2

The value of the field components

The values of H_z are given in this appendix for the different regions of integration mentioned in section 2. The values of E_x and E_y could very easily be found from the Maxwell's equations

$$E_x = \frac{j\omega\mu}{\beta^2 - \omega^2\,\varepsilon\mu}\,\frac{\partial H_z}{\partial z}$$

$$E_y = -\,\frac{j\omega\mu}{\beta^2 - \omega^2\,\varepsilon\mu}\,\frac{\partial H_z}{\partial x}\,.$$

The amplitudes of the fields of guides a and b are labeled A_0 and B_0, respectively. It is assumed that $A_0 = B_0$. The positions of the various points delimiting the integration zones are given as follows:

$$C = -\frac{c}{2\cos\theta}$$

$$B = \frac{c}{2\cos\theta}$$

$$G = \frac{c}{2}\,\sin\theta\,(1 + \tan\theta)$$

$$E = G - \frac{c}{2}$$

$$F = G + \frac{c}{2}$$

$$H = B + \frac{c}{\sin\theta}$$

The values of various lengths appearing in the expressions for the field components are given as follows:

$$IG = \frac{G}{\tan\theta}$$

$$GJ = d_0 + \frac{c}{2} - IG$$

$$OA = \frac{c}{2}\tan\theta$$

$$OI = \frac{IG}{\cos\theta}$$

It is to be noted that point G is situated at the intersection of the x-axis and of the line joining the centers of the guides. d_0 is the distance between the two guides when they are parallel. The transformation from the x-y coordinates to the x'-y' coordinates is given as follows:

$$\begin{pmatrix} y' \\ x' \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} y \\ x \end{pmatrix}$$

1. Rotation from 0° to 90°

Guide a

Region 8: $H_{az} = A_0 \cdot \cos\left(-k_x \frac{c}{2}\right) \cdot \sin\left(k_y \frac{c}{2}\right) \cdot \exp\left[\alpha(x - E)\right] \cdot \exp\left[-\alpha(y + GJ)\right]$

Region 4: $H_{az} = A_0 \cdot \cos[(k_x(x - G)] \cdot \sin\left(k_y \frac{c}{2}\right) \cdot \exp[-\alpha(y + GJ)]$

Region 6: $H_{az} = A_0 \cdot \cos\left(k_x \frac{c}{2}\right) \cdot \sin\left(k_y \frac{c}{2}\right) \cdot \exp\left[-\alpha(x - F)\right] \cdot \exp\left[-\alpha(y + GJ)\right]$

Guide b

Region 9: $H_{bz} = B_0 \cdot \cos\left(-k_x \frac{c}{2}\right) \cdot \sin\left(-k_y \frac{c}{2}\right) \cdot \exp\left[\alpha\left(x' + \frac{c}{2}\right)\right] \cdot \exp\left[\alpha(y' - OA)\right]$

Region 5: $H_{bz} = B_0 \cdot \cos\left(k_x x'\right) \cdot \sin\left(-k_y \frac{c}{2}\right) \cdot \exp\left[\alpha(y' - OA)\right]$

Region 2: $H_{bz} = B_0 \cdot \cos\left(k_x \frac{c}{2}\right) \cdot \sin\left[(k_y\,(y' - OI)\right] \cdot \exp\left[-\alpha\left(x' - \frac{c}{2}\right)\right]$

Region 6: $H_{bz} = B_0 \cdot \cos\left(k_x \frac{c}{2}\right) \cdot \sin\left(k_y \frac{c}{2}\right) \cdot \exp\left[-\alpha\left(x' - \frac{c}{2}\right)\right] \cdot \exp\left[-\alpha(y' - OA - c)\right]$

2. Rotation from 90° to 180°

Guide a

Regions 8, 4 and 6: as above.

Guide b

Region 7: $H_{bz} = B_0 \cdot \cos\left(k_x \frac{c}{2}\right) \cdot \sin\left(-k_y \frac{c}{2}\right) \cdot \exp\left[-\alpha\left(x' - \frac{c}{2}\right)\right] \cdot \exp\left[\alpha(y' + OA + c)\right]$

Region 2: $H_{bz} = B_0 \cdot \cos\left(k_x \frac{c}{2}\right) \cdot \sin\left[k_y\left(y' + \frac{c}{2} + OA\right)\right] \cdot \exp\left[-\alpha\left(x' - \frac{c}{2}\right)\right]$

Region 5: $H_{bz} = B_0 \cdot \cos\left(k_x x'\right) \cdot \sin\left(k_y \frac{c}{2}\right) \cdot \exp\left[-\alpha(y' + OA)\right]$

Region 8: $H_{bz} = B_0 \cdot \cos\left(-k_x \frac{c}{2}\right) \cdot \sin\left(k_y \frac{c}{2}\right) \cdot \exp\left[\alpha\left(x' + \frac{c}{2}\right)\right] \cdot \exp\left[-\alpha(y' + OA)\right]$

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Figure Captions

Figure 1: The mapping of the nine regions into which the x-y plane is divided.

Figure 2: The geometry of the two rectangular dielectric waveguides and the path of integration
a) the inclination angle $\theta = 0°$
b) the inclination angle θ is different from zero.

Figure 3: The subdivision of the x-axis into zones for $0° \leq \theta \leq 90°$. $a_1 = a_2 = b_1 = b_2 = c$.

Figure 4: The subdivision of the x-axis into zones for $90° \leq \theta \leq 180°$. $a_1 = a_2 = b_1 = b_2 = c$.

Figure 5: The integrand of C as a function of the distance along the x-axis for various inclination angles θ.

Figure 6: The coupling coefficient $\Delta\beta = C/P$ as a function of the inclination angle θ.

Figure 7: The subdivision of the x-axis into zones for guides parallel coordinate axes.
$a_1 = a_2 = b_1 = b_2 = c$.

Figure 8: The subdivision of the x-axis into zones for guides perpendicular coordinate axes.
$a_1 = a_2 = b_1 = b_2 = c$.

Figure 9: The integrand of C as a function of the distance along the x-axis for parallel (a) and perpendicular (b) cases.

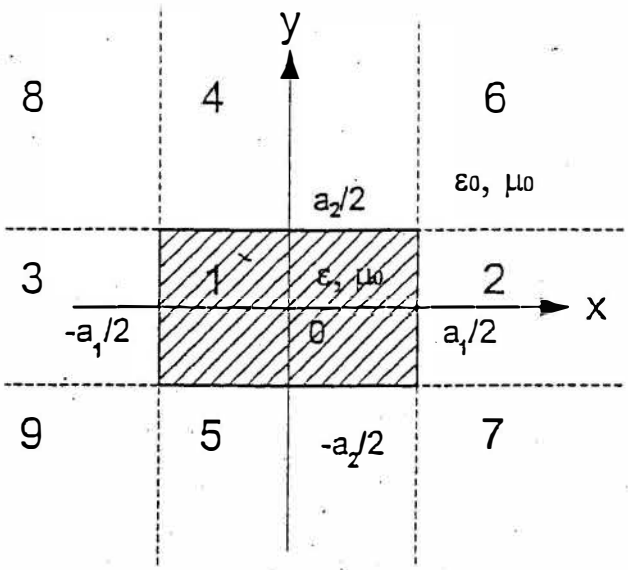
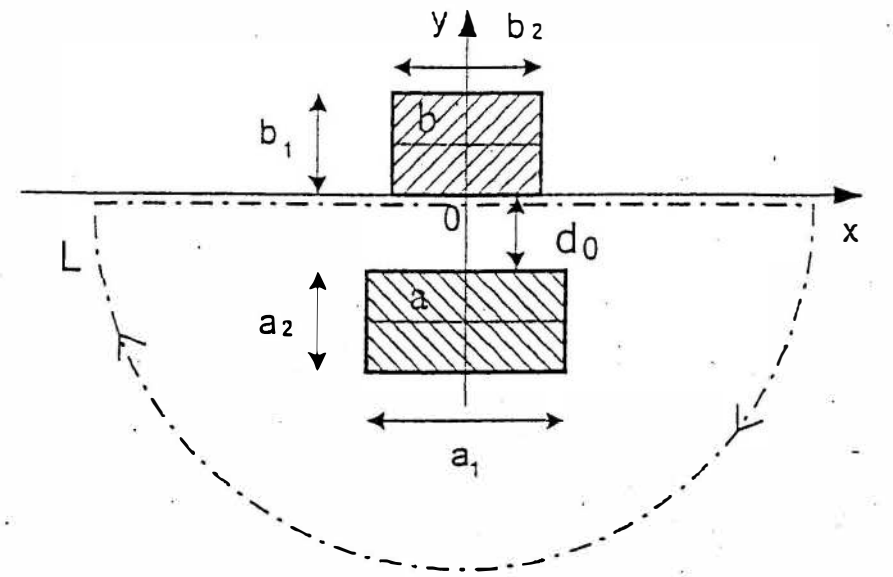
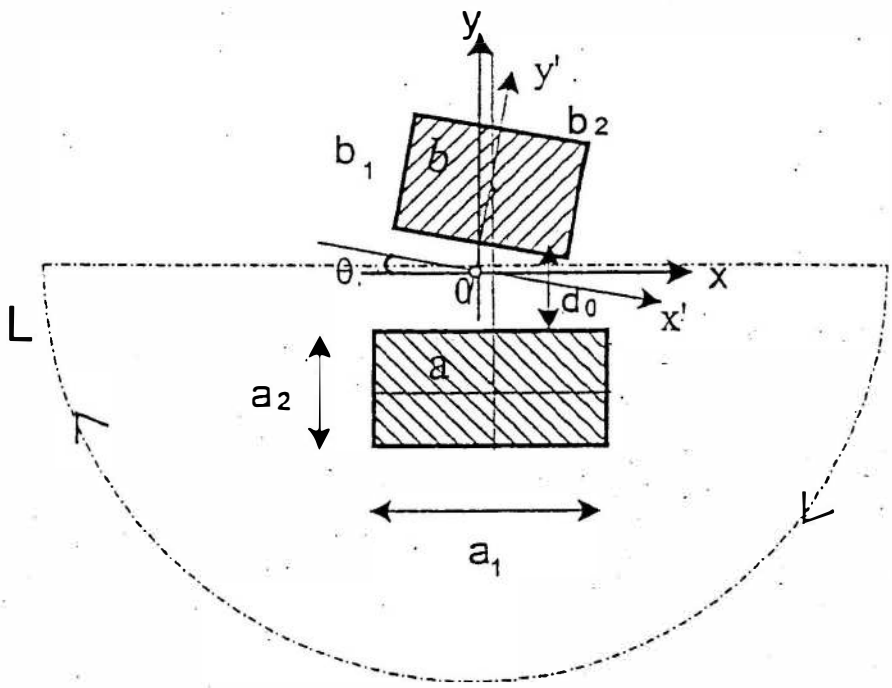


Figure 1



a



b

Figure 2

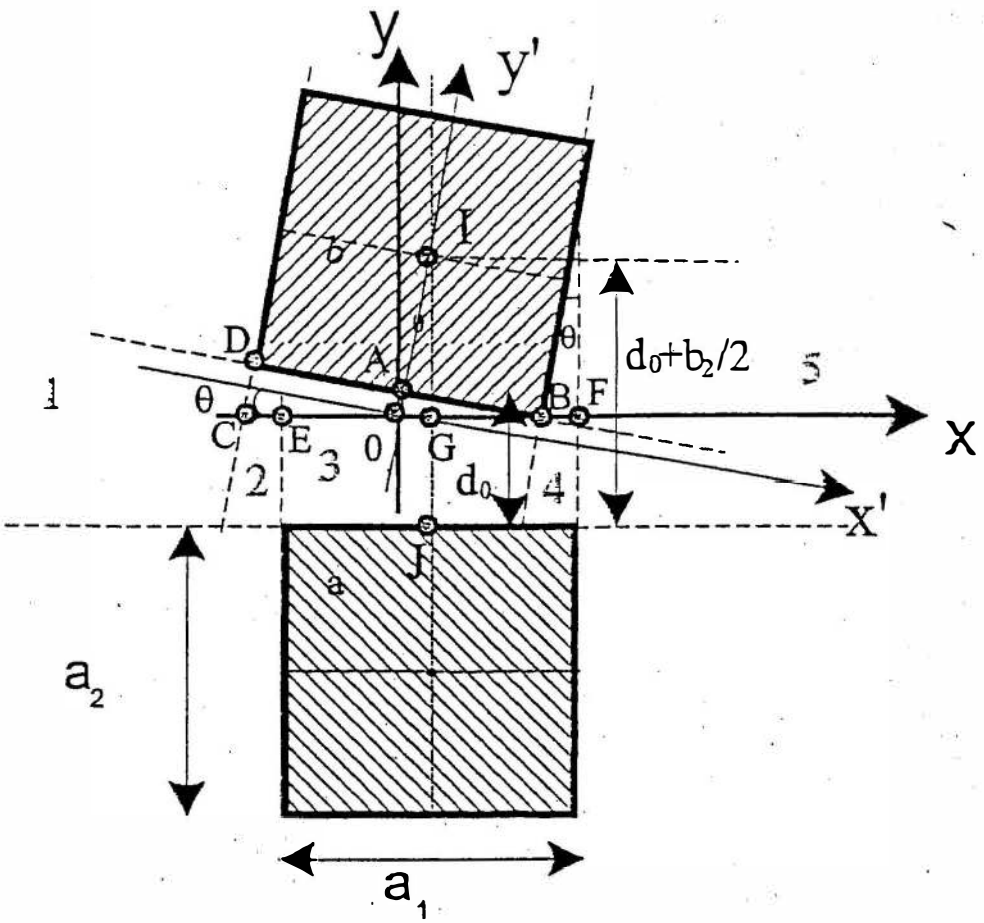


Figure 3

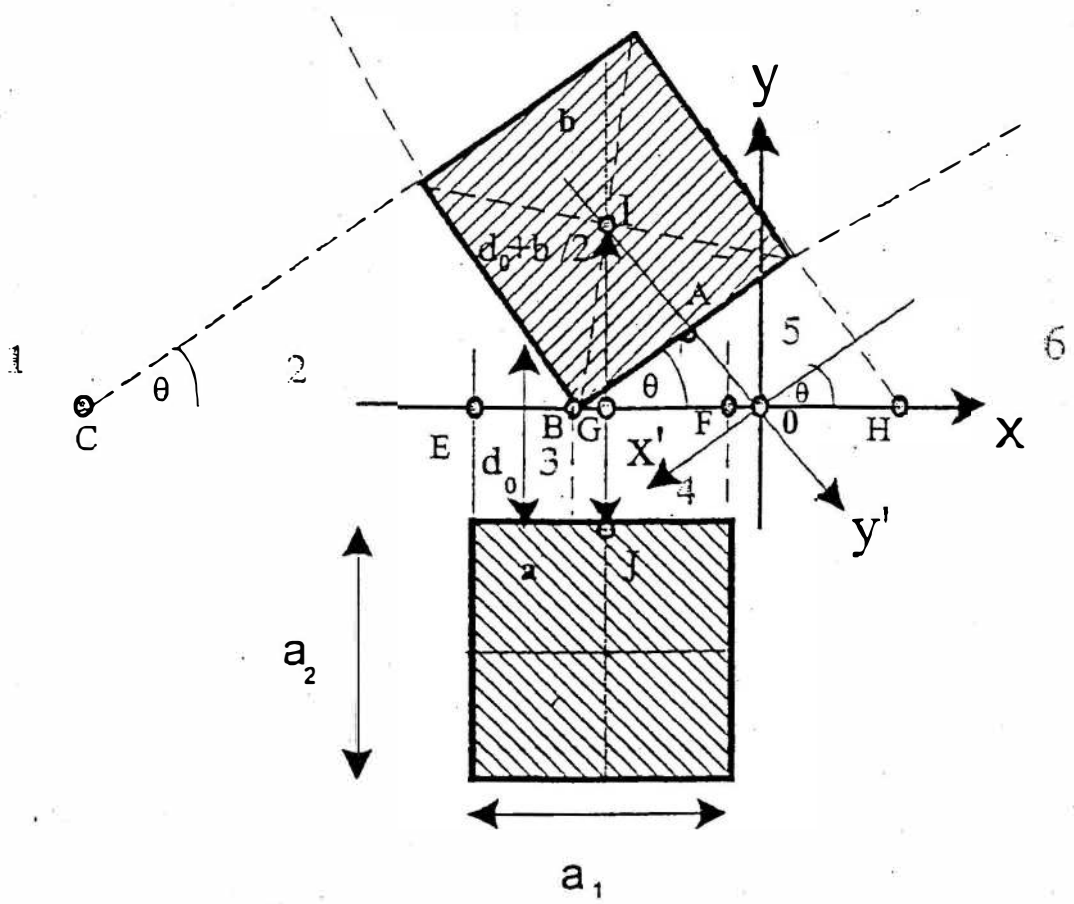


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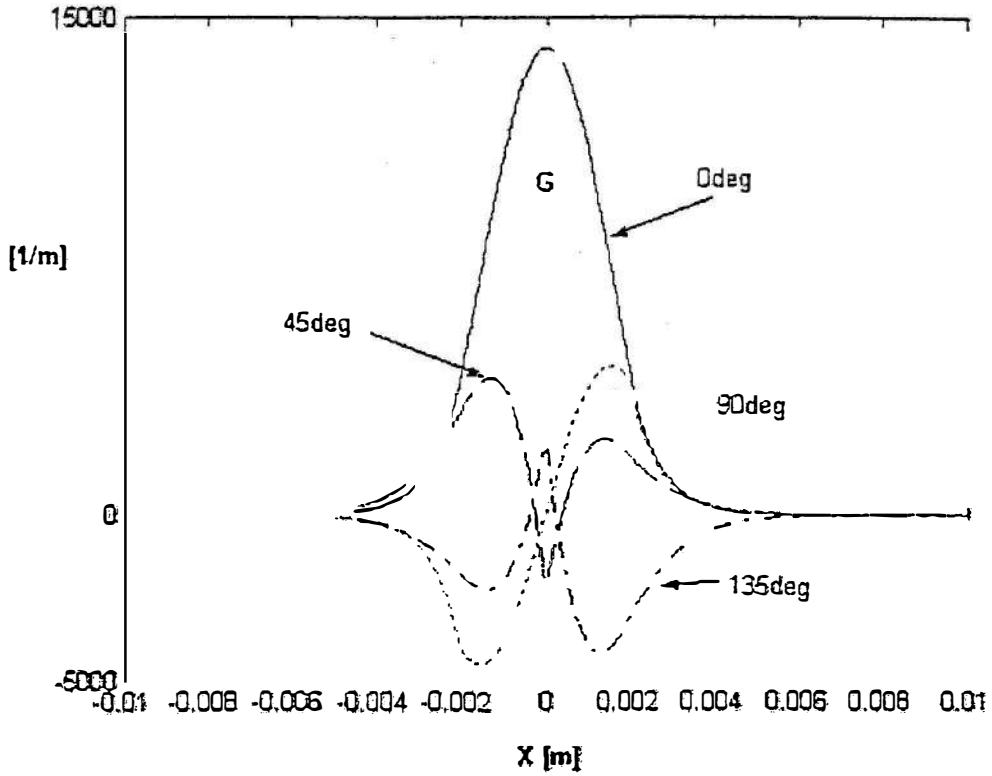

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[1/m]
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0deg
45deg
90deg
0
135deg
-5000
-0.01
-0.008
-0.006
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-0.002
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0.002
0.004
0.006
0.008
0.01
X [m]

Figure 5

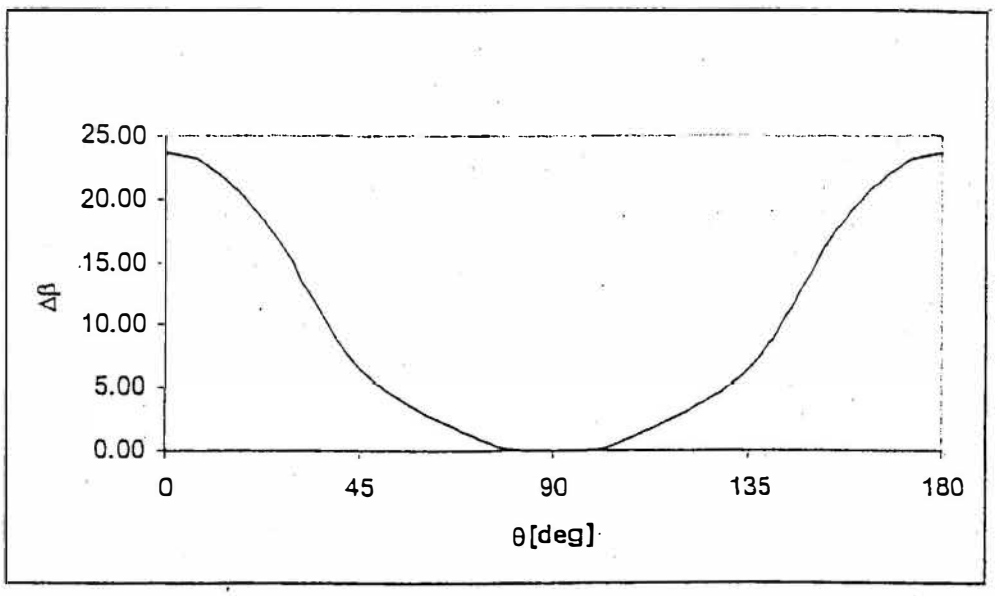


Figure 6

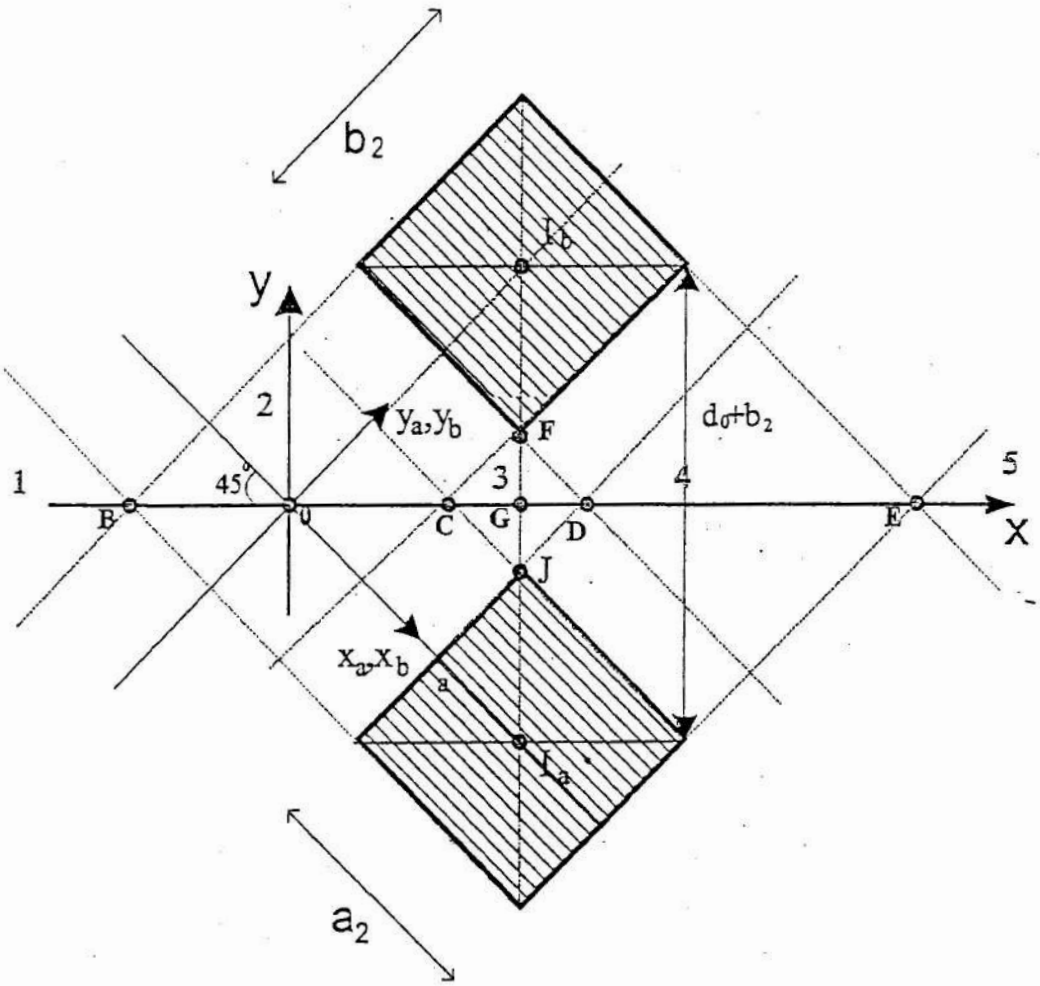


Figure 7

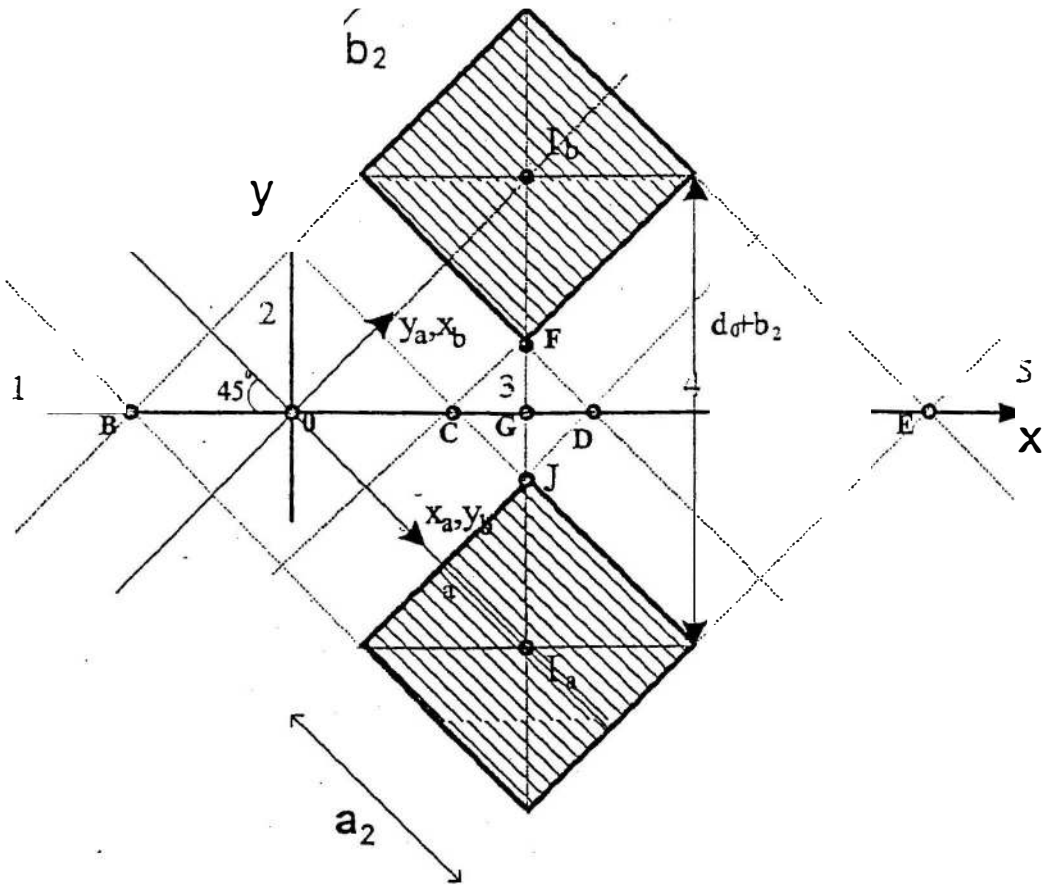


Figure 8

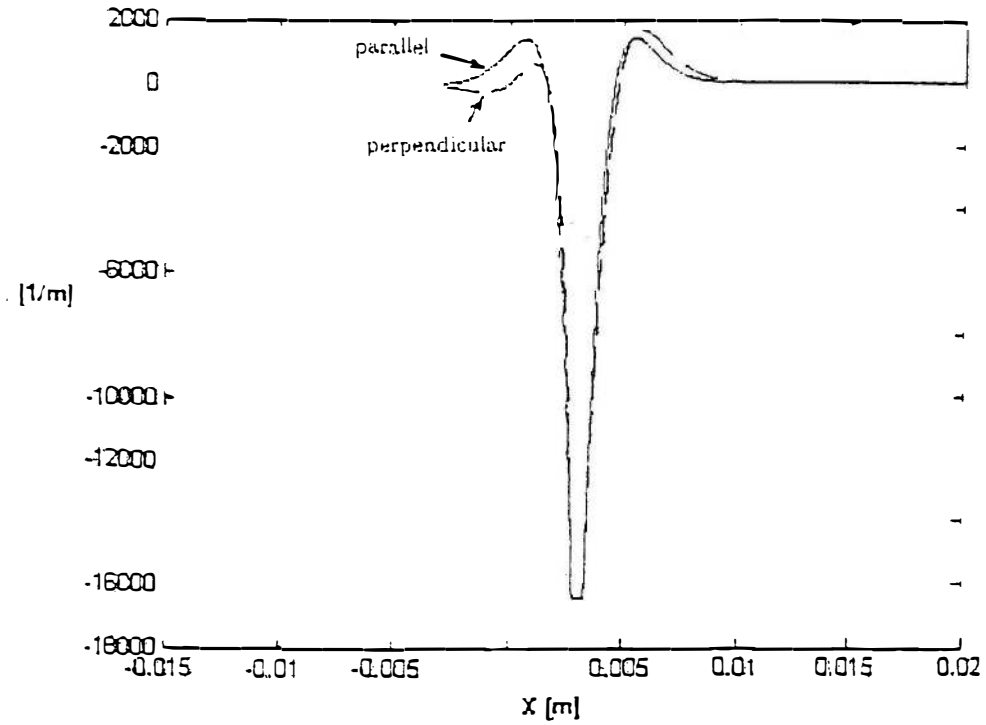

2000
0
-2000
-6000
-10000
-12000
-16000
-18000
[1/m]
parallel
perpendicular
-0.015
-0.01
-0.005
0.005
0.01
0.015
0.02
X [m]

Figure 9

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